

EDMD audit of relaxation modes in the controlled three-mode NLS system

Numerical validation report

7 September 2026

Abstract

We applied Extended Dynamic Mode Decomposition (EDMD) to four existing, projection-free Cartesian trajectory ensembles for the three-mode chain. The state could be reconstructed from every stored snapshot, so no new dynamics were simulated. The constant mode, Gram conditioning, cutoff sensitivity, dictionary, lag, timestep, and whole-trajectory bootstrap checks were fixed before spectra were inspected. In the driven system only one nontrivial rate passes every gate, $\lambda \simeq -0.95$. Two faster candidates cannot be statistically separated, and the known oscillation near $\text{Im } \lambda \simeq 5.3$ fails lag convergence. The calculation therefore does not establish several distinct driven slow modes. It supports one common leading mode with observable-dependent contamination from unresolved faster modes. At equilibrium, real rates near -0.95 and -2.85 pass all gates at both timesteps.

1 Question and data

The controlled autocorrelation study found, at the fine timestep in the driven case, plateau rates -0.9351 for I_2 , -1.0028 for $I_1 + I_3$, -1.0921 for $\cos \theta_3$, and -1.1218 for $\cos \theta_1 + \cos \theta_3$. EDMD asks whether these are mixtures of several generator modes rather than estimates of one unresolved gap.

The input consists of 64 independent trajectories for each bath condition $(T_1, T_3) = (2, 8)$ and $(5, 5)$ and each integration timestep 10^{-3} and 2.5×10^{-4} . Each trajectory has burn-in 500, stationary duration 1000, and 100001 snapshots at spacing 0.01. The simulator used double-precision state, a fixed timestep, standard trigonometry, and no action floor or projection; snapshots were written as float32 observables.

No rerun was required. The full reduced state is algebraically reconstructed:

$$I_1 = \frac{1}{2}[(I_1 + I_3) + (I_1 - I_3)], \quad I_3 = \frac{1}{2}[(I_1 + I_3) - (I_1 - I_3)],$$

with I_2 saved directly and $\theta_r = \text{atan2}(\sin \theta_r, \cos \theta_r)$.

2 Frozen EDMD estimator

Actions use standardized logarithms $z_j = (\log \max(I_j, 10^{-12}) - \overline{\log I_j})/s_j$. Action monomials have total degree at most d . The angular factor is the real Fourier basis on the two-torus with $|m_1|, |m_3| \leq M$. Their tensor product gives dictionaries D1 $(d, M) = (2, 1)$, D2 $(2, 2)$, and D3 $(3, 2)$, with $K = 90, 250, 500$ respectively; the constant is always included.

Each stream contributes 2048 fixed, evenly spaced origins, giving 131072 pairs per case. At lags $\tau = 0.10, 0.25, 0.50, 1.00$ we estimate

$$A_{ij} = \langle \psi_i(X_s) \psi_j(X_s) \rangle, \quad B_{ij} = \langle \psi_i(X_s) \psi_j(X_{s+\tau}) \rangle, \quad K_\tau = A^+ B.$$

The primary Gram-eigenvalue cutoff is 10^{-10} relative to the largest eigenvalue, with fixed sensitivity checks at 10^{-8} and 10^{-12} . Rates use the principal logarithm $\lambda = \log \mu / \tau$. Uncertainty is from 500 resamples of the 64 complete trajectories.

The complete predeclared gates are in `PROTOCOL.md`. Briefly, a mode must be stable across all three dictionaries, all four lags, and both timesteps; have at least 90% successful bootstrap matches; and have a real-part interval separated from neighbouring reportable candidates.

3 Mandatory convergence checks

3.1 Trivial mode, conditioning, and cutoff

All full-sample fits recover the constant eigenfunction with $|\mu - 1| < 1.2 \times 10^{-14}$. D3 retains all 500 Gram directions. Its condition number ranges from 4.27×10^3 to 1.41×10^4 across the four datasets. Because the smallest retained

direction remains above 10^{-8} of the largest, all three fixed cutoffs retain the same 500 directions and give identical printed spectra. These controls pass.

3.2 Leading-rate convergence

Table 1 gives the leading nontrivial rate before combining all gates. The D1–D3 drift is appreciable but lies inside the frozen 20% tolerance; lag and timestep gates also pass.

Table 1: Leading nontrivial real rate. Brackets are 95% whole-stream bootstrap intervals for D3 at $\tau = 0.5$.

case	D1	D2	D3 [95% CI]	D3 lags 0.1 $\rightarrow$ 1.0
driven, 10^{-3}	-1.1053	-1.0904	-0.9402 [-0.9546,-0.9208]	-0.9779,-0.9619,-0.9402,-0.9218
driven, $2.5 \cdot 10^{-4}$	-1.1250	-1.1118	-0.9632 [-0.9800,-0.9419]	-1.0558,-0.9870,-0.9632,-0.9451
equil., 10^{-3}	-1.1357	-1.1232	-0.9551 [-0.9724,-0.9328]	-1.0718,-1.0164,-0.9551,-0.9335
equil., $2.5 \cdot 10^{-4}$	-1.1121	-1.0980	-0.9408 [-0.9548,-0.9230]	-1.0518,-0.9748,-0.9408,-0.9212

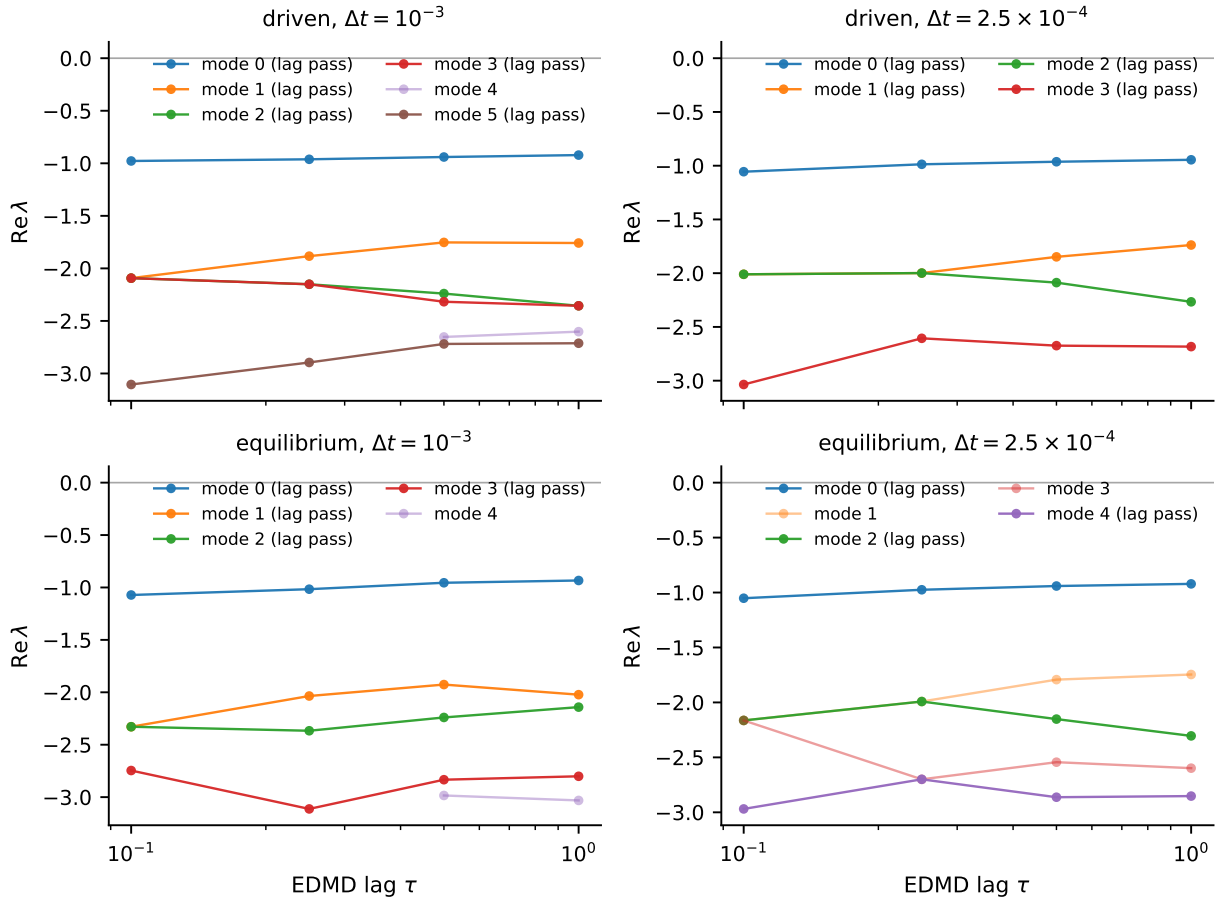


Figure 1: Raw lag dependence of the first frozen reference candidates. Faded curves fail that candidate’s lag gate; a lag pass alone does not imply final reportability.

3.3 Modes that survive every gate

The driven candidates near -1.8 and -2.1 each pass the point-estimate dictionary/lag/timestep tolerances but fail separation: at the fine timestep their intervals are $[-1.991, -1.701]$ and $[-2.303, -1.779]$, which overlap. At the coarse timestep the corresponding intervals are $[-1.971, -1.533]$ and $[-2.542, -1.895]$. They cannot be claimed as two resolved eigenvalues.

Table 2: Final reportable modes. Rates absent from this table remain in the raw spectra and failed at least one frozen gate.

bath	timestep	Re λ	95% CI	Im λ
driven	10^{-3}	-0.940174	[-0.954628,-0.920824]	0
driven	$2.5 \cdot 10^{-4}$	-0.963191	[-0.980007,-0.941942]	0
equilibrium	10^{-3}	-0.955149	[-0.972388,-0.932803]	0
equilibrium	$2.5 \cdot 10^{-4}$	-0.940758	[-0.954830,-0.922965]	0
equilibrium	10^{-3}	-2.833818	[-2.946075,-2.705420]	0
equilibrium	$2.5 \cdot 10^{-4}$	-2.862538	[-2.931237,-2.741557]	0

3.4 Known oscillatory control

At $\tau = 0.5$, a complex candidate can be found with imaginary part close to 5.3 in every dataset. However, its real part and, at long lag, its imaginary part drift beyond the frozen tolerance. For example, the driven fine-step candidate is $-7.273 + 5.317i$ at $\tau = 0.5$, while its D3 lag sequence is $-7.092 + 4.940i$, $-7.476 + 5.666i$, $-7.273 + 5.317i$, and $-5.016 + 3.142i$. The last value is the principal-log Nyquist boundary π/τ , not recovery of the known frequency. All four cases fail the lag gate. Thus EDMD does not recover the independent 5.3 known answer, and the dictionary is inadequate for the complete spectrum.

4 Observable projections and the key question

For the driven fine-step fit, the signed contribution of the one reportable slow mode to the reconstructed zero-lag covariance is 0.490 for I_2 , 0.283 for $I_1 + I_3$, 0.123 for $\cos \theta_3$, and 0.224 for the cosine sum. The remaining reconstructed covariance lies in faster candidates, whose weights differ by observable but whose rates do not pass all separation gates.

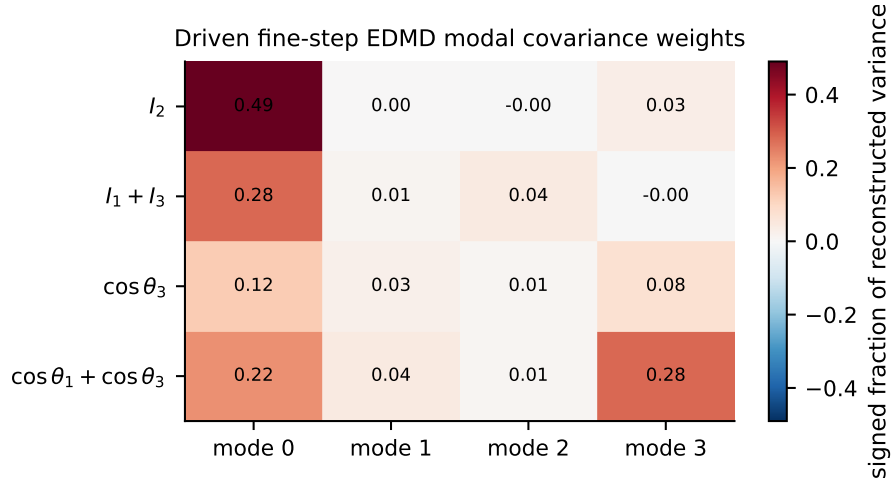


Figure 2: Signed modal covariance weights for the first four driven fine-step reference candidates. Only mode 0 is finally reportable; the other columns are diagnostic and must not be read as validated eigenvalues.

This pattern is consistent with the autocorrelation plateaus being finite-time mixtures: I_2 has the largest weight on the slow mode and gave the least negative plateau, whereas the angular observables carry more unresolved faster weight and gave more negative plateaus. It does *not* establish the stronger hypothesis of several distinct, resolved driven slow modes. The strict answer to the key question is therefore negative at the present dictionary and data resolution.

5 Eigenfunction slices

Figure 3 shows the real parts of all fine-timestep modes that survive the complete gates: the driven and equilibrium leading modes and the additional equilibrium mode near -2.86 . Each row has its own colour scale,

so amplitudes should not be compared between rows. Numerical values on the full 81×81 grids are saved in `analysis/reportable_eigenfunction_slices.csv`.

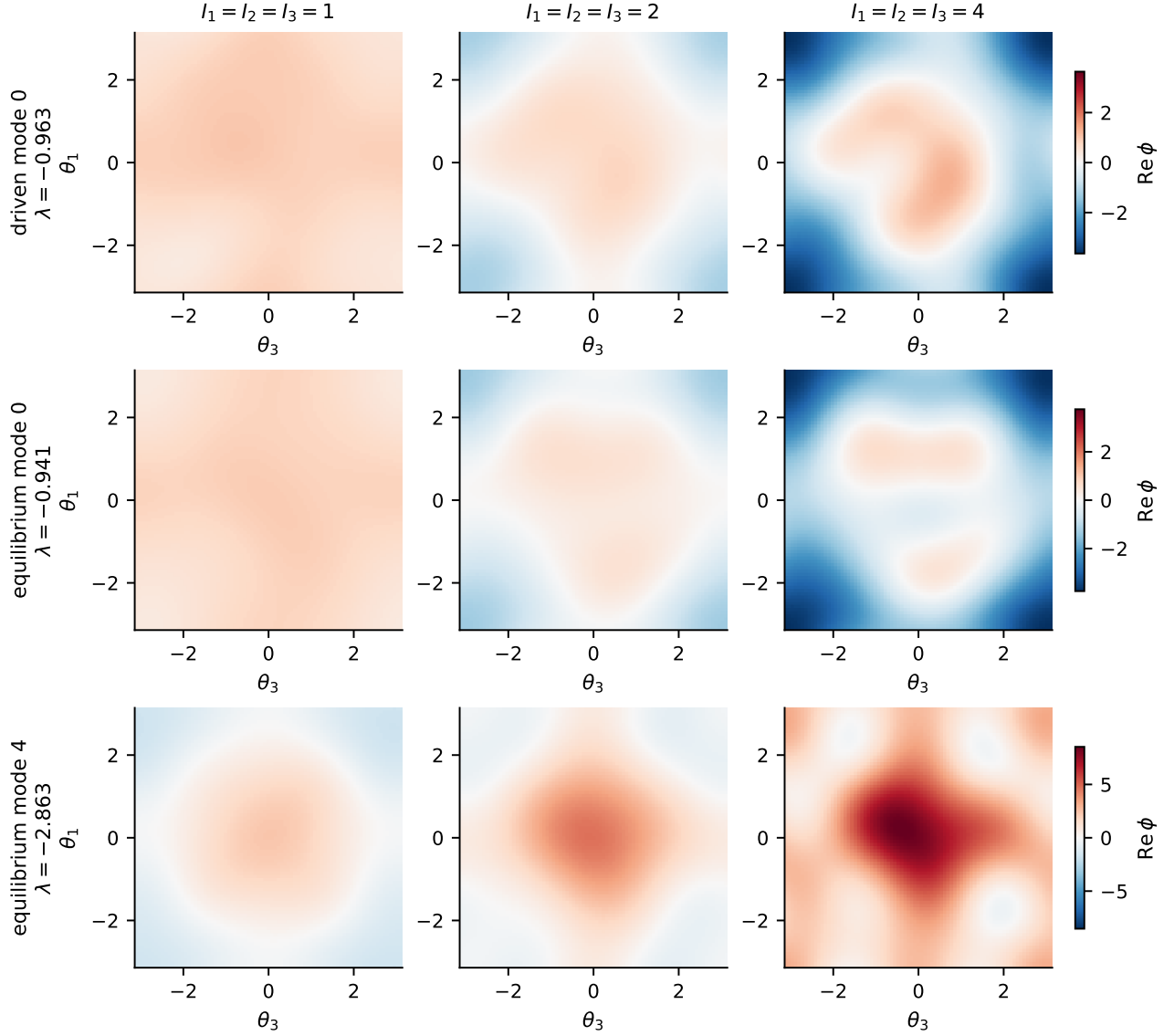


Figure 3: EDMD eigenfunction slices in the (θ_1, θ_3) plane at equal actions 1, 2, and 4.

6 Conclusion and claim boundary

The EDMD computation resolves a robust leading real mode near -0.95 in both bath conditions and both timesteps. It also resolves a faster equilibrium mode near -2.85 . In the driven case, no second nontrivial rate is both converged and statistically separated. The known oscillatory mode is not recovered under the lag gate. Accordingly, the defensible claim is:

EDMD independently identifies a common leading relaxation rate near -0.95 and shows strongly observable-dependent modal weights, consistent with finite-window contamination of single-observable autocorrelation rates. It does not yet resolve a multi-mode driven spectrum or the known oscillatory pair.

This is not an exact generator-spectrum proof and not a unique spectral-gap theorem.

7 Provenance and raw outputs

The trajectory simulator source SHA-256 is 58c871882f1180701a7aae637a8f4be113315a72fb9d7815562f5fc90faf7976; the binary SHA-256 is f7cd820bb8f716a8790f688db060259da5657e4f20fa09100546e66efb696b3f. The production launch commit is 899a5d345a98e4c8d5605a6a5a166a08bb86adcd. Base seeds are 2026090801–2026090804 in case order; all 64 deterministic stream seeds per case are in the original `metadata.csv` files. The EDMD protocol was frozen before spectrum computation, and the analysis source run had SHA-256 d4c53270ed902cbcf6692dcc64d7c1f423e40189504d03703c6e4f8cd50cc6ef.

The exact command was

```
python3 run_edmd.py --input ../spectral_gap_controlled_2026-09-07/raw \  
--output analysis
```

followed by the frozen gate summarizer and figure/audit scripts recorded in `provenance/commands.txt`. All A and B matrices are compressed CSV files under `analysis/matrices`; all eigenvalues, 500 replicate-level bootstrap matches, Gram diagnostics, observable weights, convergence gates, and eigenfunction grids are CSV files under `analysis`. The automated integrity audit passed 42/42 checks. SHA-256 values are in `provenance/analysis_outputs.sha256`.